

Refrigeration Heat Load Calculation

Legacy v1 project

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Customer	Demo client
Site	Cairo, Egypt, Dammam, Saudi Arabia
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Contents

- | | |
|------------------------------------|--------------------------------------|
| 1. Project information | 8. Final refrigeration load |
| 2. Design criteria | 9. Tunnel / blast freezers |
| 3. Engineering assumptions | 10. Machinery room ventilation |
| 4. Input data | 11. Engineering notes and validation |
| 5. Detailed heat load calculations | 12. Standards and references |
| 6. Load summary | 13. Revision history |
| 7. Design allowances | 14. Prepared / Checked / Approved |

1. Project information

Project name	Legacy v1 project
Project number	PRJ-2026-0001
Customer	Demo client
End user	—
Consultant	—
Contractor	—
Site	Cairo, Egypt
City / country	Dammam, Saudi Arabia
Project type	—
Status	Draft
Created	2026-09-25 by QA Engineer
Modified	2026-09-25 by QA Engineer

2. Design criteria

Outdoor design dry-bulb	47.0 °C
Coincident relative humidity	16 %
Site altitude	10 m
Ground / under-floor temperature	12.0 °C
Default safety factor	10 %
Refrigerant / system	R717 (Ammonia)
Weather data source	Dammam, Saudi Arabia — 0.4 % DB 47.0 °C / MCWB 25.0 °C (QA test values)
Heat loss to colder surroundings	Credited

Room	Application	L×W×H [m]	T [°C]	RH [%]	Run [h/day]	TD [K]	Safety [%]
Frozen store FS-01	Frozen store (−18...−30 °C)	40.0 × 25.0 × 11.0	−25.0	85	20	7.0	10
Fruit store CS-01	Fruit & vegetable store (high RH)	30.0 × 20.0 × 9.0	1.0	90	18	4.5	10
Anteroom / dock AR-01	Anteroom / loading dock	40.0 × 10.0 × 7.0	5.0	75	18	8.0	10

3. Engineering assumptions

Scope	Item	Value	Type	Basis
Project	Outdoor design condition	47 °C / 16.4 % RH, 10 m	User-selected	Dammam, Saudi Arabia — 0.4 % DB 47.0 °C / MCWB 25.0 °C (QA test values)
Project	Ground temperature	12 °C	User-selected	Floor on ground / heated slab
Method	Surface film coefficients	h _i 9.4, h _o 34 (outdoor), 9.4 (adjacent) W/m²K	Default	Still air inside; wind outside
Method	Product properties	Siebel equations from water content; latent heat assumes all water freezes	Default	ASHRAE Thermal Properties of Foods
Method	Heat loss to colder surroundings	Credited (reduces load)	Default	Project option (Gate 3 G3-3)
Method	Occupancy heat	272 – 6-t W per person	Default	ASHRAE / Dossat
Frozen store FS-01	Run time	20 h/day	Default	Preset for Frozen store (−18...−30 °C)
Frozen store FS-01	Evaporator TD	7 K	Default	
Frozen store FS-01	Safety factor	10 %	Default	
Frozen store FS-01	Product load capacity basis	Daily: Q × 24/t _{pull} ÷ run time	Default	Room option (Gate 3 G3-2)
Frozen store FS-01	Poultry (chicken, whole) water content	74 %	Database	Product database (typical)
Frozen store FS-01	Poultry (chicken, whole) freezing point	−2.8 °C	Database	Product database (typical)
Frozen store FS-01	Packaging	5 % Corrugated cardboard (c _p 1.34)	User-selected	
Frozen store FS-01	Dock door to anteroom protection	PVC strip curtain (E = 0.85)	Default	Door protection effectiveness
Frozen store FS-01	Dock door to anteroom flow factor D _f	auto (0.8 / 1.1)	Default	
Frozen store FS-01	Evaporator fans	8 % allowance	Default	Estimate before coil selection

Scope	Item	Value	Type	Basis
Frozen store FS-01	Defrost heat to room	30 %	User-selected	
Fruit store CS-01	Run time	18 h/day	Default	Preset for Fruit & vegetable store (high RH)
Fruit store CS-01	Evaporator TD	4.5 K	Default	
Fruit store CS-01	Safety factor	10 %	Default	
Fruit store CS-01	Product load capacity basis	Daily: $Q \times 24/t_{\text{pull}} \div \text{run time}$	Default	Room option (Gate 3 G3-2)
Fruit store CS-01	Apples water content	84 %	Database	Product database (typical)
Fruit store CS-01	Apples freezing point	-1.1 °C	Database	Product database (typical)
Fruit store CS-01	Apples respiration	10 W/t	Database	Product database (typical)
Fruit store CS-01	Packaging	10 % Wooden pallets / crates (c_p 2.5)	User-selected	
Fruit store CS-01	Loading door protection	PVC strip curtain ($E = 0.85$)	Default	Door protection effectiveness
Fruit store CS-01	Loading door flow factor D_f	auto (0.8 / 1.1)	Default	
Fruit store CS-01	Evaporator fans	8 % allowance	Default	Estimate before coil selection
Anteroom / dock AR-01	Run time	18 h/day	Default	Preset for Anteroom / loading dock
Anteroom / dock AR-01	Evaporator TD	8 K	Default	
Anteroom / dock AR-01	Safety factor	10 %	Default	
Anteroom / dock AR-01	Product load capacity basis	Daily: $Q \times 24/t_{\text{pull}} \div \text{run time}$	Default	Room option (Gate 3 G3-2)
Anteroom / dock AR-01	Dock levellers ×6 (sealed) protection	$E = 0.9$	Overridden	Door protection effectiveness
Anteroom / dock AR-01	Dock levellers ×6 (sealed) flow factor D_f	auto (0.8 / 1.1)	Default	
Anteroom / dock AR-01	Evaporator fans	8 % allowance	Default	Estimate before coil selection
Blast freezer BF-01	Freezing-time basis	Pham method (ASHRAE)	Default	Tunnel option
Blast freezer BF-01	Surface coefficient / packaging resistance	20 W/m²K / 0.01 m²K/W	User-selected	From air velocity, supplier or ASHRAE data
Blast freezer BF-01	Frozen product k / ρ	1.4 W/mK / 1000 kg/m³	User-selected	Typical values — verify (ASHRAE Thermal Properties of Foods)
Blast freezer BF-01	Loss & safety	10 % ($Q \div (1 - x)$)	Default	Workbook method

4. Input data

Frozen store FS-01

Envelope

Surface	Adjacent	T adj [°C]	Insulation	Thk [mm]	Sun	Area ovr	U ovr
Wall 1 (length)	ambient	design DB	PUR	175	none	—	—
Wall 2 (width)	ambient	design DB	PUR	175	none	—	—
Wall 3 (length)	ambient	design DB	PUR	175	none	—	—
Wall 4 (width)	ambient	design DB	PUR	175	none	—	—
Ceiling / roof	ambient	design DB	PUR	175	none	—	—
Floor	ground		XPS	150	—	—	—

Product

Product	kg/day	T in → out [°C]	Pull-down [h]	x_w [%]	T_f [°C]	Pack.	Stored [kg]	Resp. [W/t]
Poultry (chicken, whole)	40,000	-15 → -25	24	74	-2.8	5 % cardboard	0	0

Infiltration

Door	W×H [m]	Opens to	Passages/day	Open-close [s]	Stand open [min/day]	Protection	D_f
Dock door to anteroom	2.4 × 3	5 °C / 75 %	150	15	0	strip	auto

Internal & equipment

People 2 × 8 h · Lighting 8 W/m² × 12 h · Forklifts 2 × 7 kW × 12 h · Other 0 kW × 0 h · Fans 8 % · Defrost 60 kW × 3 × 30 min × 30 % · Ventilation 0 m³/h × 24 h

Fruit store CS-01

Envelope

Surface	Adjacent	T adj [°C]	Insulation	Thk [mm]	Sun	Area ovr	U ovr
Wall 1 (length)	ambient	design DB	PUR	100	none	—	—
Wall 2 (width)	ambient	design DB	PUR	100	none	—	—
Wall 3 (length)	ambient	design DB	PUR	100	none	—	—

Surface	Adjacent	T adj [°C]	Insulation	Thk [mm]	Sun	Area ovr	U ovr
Wall 4 (width)	ambient	design DB	PUR	100	none	—	—
Ceiling / roof	ambient	design DB	PUR	100	none	—	—
Floor	ground		XPS	100	—	—	—

Product

Product	kg/day	T in → out [°C]	Pull-down [h]	x_w [%]	T_f [°C]	Pack.	Stored [kg]	Resp. [W/t]
Apples	60,000	22 → 1	24	84	-1.1	10 % wood	800,000	10

Infiltration

Door	W×H [m]	Opens to	Passages/day	Open-close [s]	Stand open [min/day]	Protection	D_f
Loading door	2.4 × 3	8 °C / 70 %	80	20	10	strip	auto

Internal & equipment

People $2 \times 8 \text{ h}$ · Lighting $8 \text{ W/m}^2 \times 12 \text{ h}$ · Forklifts $1 \times 6 \text{ kW} \times 4 \text{ h}$ · Other $0 \text{ kW} \times 0 \text{ h}$ · Fans 8 % · Defrost $0 \text{ kW} \times 4 \times 30 \text{ min} \times 30 \%$ · Ventilation $0 \text{ m}^3/\text{h} \times 24 \text{ h}$

Anteroom / dock AR-01**Envelope**

Surface	Adjacent	T adj [°C]	Insulation	Thk [mm]	Sun	Area ovr	U ovr
Wall 1 (length)	ambient	design DB	PUR	80	none	—	—
Wall 2 (width)	ambient	design DB	PUR	80	none	—	—
Wall 3 (length)	ambient	design DB	PUR	80	none	—	—
Wall 4 (width)	ambient	design DB	PUR	80	none	—	—
Ceiling / roof	ambient	design DB	PUR	80	none	—	—
Floor	ground		XPS	100	—	—	—

Product

No product.

Infiltration

Door	W×H [m]	Opens to	Passages/day	Open-close [s]	Stand open [min/day]	Protection	D_f
Dock levellers ×6 (sealed)	2.8 × 3	Outdoor	120	30	60	E=0.9	auto

Internal & equipment

People $8 \times 16 \text{ h}$ · Lighting $8 \text{ W/m}^2 \times 12 \text{ h}$ · Forklifts $3 \times 6 \text{ kW} \times 12 \text{ h}$ · Other $0 \text{ kW} \times 0 \text{ h}$ · Fans 8 % · Defrost $0 \text{ kW} \times 4 \times 30 \text{ min} \times 30 \%$ · Ventilation $0 \text{ m}^3/\text{h} \times 24 \text{ h}$

5. Detailed heat load calculations

Values in SI (the calculation engine's internal units).

Frozen store FS-01

Transmission load

$$Q = U \cdot A \cdot \Delta T, U = 1 / (1/h_i + L/k + 1/h_o + R_{\text{extra}}), \Delta T = T_{\text{out}} (+ \text{sun effect}) - T_{\text{room}}$$

Surface	A [m²]	U [W/m²K]	T _{out} [°C]	ΔT [K]	Q [kW]	Q [kWh/day]
Wall 1 (length)	440.0	0.124	47.0	72.0	3.916	94.0
Wall 2 (width)	275.0	0.124	47.0	72.0	2.447	58.7
Wall 3 (length)	440.0	0.124	47.0	72.0	3.916	94.0
Wall 4 (width)	275.0	0.124	47.0	72.0	2.447	58.7
Ceiling / roof	1,000.0	0.124	47.0	72.0	8.900	213.6
Floor	1,000.0	0.192	12.0	37.0	7.107	170.6

$h_i = 9.4 \text{ W/m}^2\text{K}$; $h_o = 34 \text{ W/m}^2\text{K}$ outdoors, $9.4 \text{ W/m}^2\text{K}$ to adjacent spaces, none to ground.

$T_{\text{room}} = -25.0 \text{ °C}$.

Heat loss to colder surroundings is credited (negative loads reduce the total).

Result: **689.58 kWh/day**

Product, packaging and respiration

$$q = c_{\text{pa}}(T_1 - T_f) + h_{\text{if}} + c_{\text{pb}}(T_f - T_2); c_{\text{pa}} = 3.35 \cdot x_w + 0.84, c_{\text{pb}} = 1.26 \cdot x_w + 0.84, h_{\text{if}} = 334 \cdot x_w \text{ [kJ/kg]} \text{ (unless entered)}; Q = m \cdot q / 3600 \times 24 / t_{\text{pull}} \div \text{CRF};$$

$$Q_{\text{pack}} = m_{\text{pack}} \cdot c_p \cdot \Delta T / 3600 \times 24 / t_{\text{pull}}; Q_{\text{resp}} = (\text{stored}[t] \times w + \text{intake}[t/\text{day}] \times w_{\text{in}}) \times 24 / 1000$$

Product	m [kg/day]	T ₁ →T ₂ [°C]	T _f [°C]	c _{pa} / c _{pb} [kJ/kg·K]	h _{if} [kJ/kg]	q [kJ/kg]	t _{pull} [h]	CRF	Product [kWh/day]	Pack. [kWh/day]	Resp. [I]
Poultry (chicken, whole)	40,000	-15.0 → -25.0	-2.8	3.32 / 1.77	247.2	17.7	24.0	1.00	196.9	7.4	

Specific and latent heats from water content (Siebel; latent heat assumes all water freezes) unless entered values are shown as overridden.

Capacity basis "daily": load concentrated into the pull-down time, expressed per 24 h, then divided by run time with the other loads.

Result: **204.38 kWh/day**

Door infiltration — Gosney & Olama

$$q = 0.221 \cdot A \cdot (h_i - h_r) \cdot \rho_r \cdot (1 - \rho_i/\rho_r)^{0.5} \cdot (g \cdot H)^{0.5} \cdot F_m, F_m = [2 / (1 + (\rho_r/\rho_i)^{(1/3)})]^{1.5}; Q = q \cdot t_{\text{open}} \cdot D_f \cdot (1 - E)$$

Door	A [m²]	h _i / h _r [kJ/kg]	ρ _i / ρ _r [kg/m³]	q open [kW]	t _{open} [h/day]	D _f	E	Q [kWh/day]	Moisture [kg/day]
Dock door to anteroom	7.20	15.2 / -24.3	1.264 / 1.421	156.01	0.625	0.80	0.85	11.7	3.96

$t_{\text{open}} = (\text{passages} \times \text{open-close time} + 60 \times \text{minutes standing open}) / 3600$.

Air properties at site pressure (ASHRAE psychrometric equations).

Result: **11.70 kWh/day**

Internal loads

$$\text{People: } n \cdot (272 - 6 \cdot t) \text{ W} \cdot \text{h}; \text{ Lighting: } \text{W/m}^2 \cdot A_{\text{floor}} \cdot \text{h}; \text{ Forklifts: } n \cdot \text{kW} \cdot \text{h}; \text{ Other: } \text{kW} \cdot \text{h}$$

Item	Inputs	Q [kWh/day]
People	2 × 422 W × 8 h	6.8
Lighting	8 W/m² × 1,000 m² × 12 h	96.0
Forklifts	2 × 7 kW × 12 h	168.0
Other equipment	0 kW × 0 h	0.0

Result: **270.75 kWh/day**

Evaporator fans and defrost

$$\text{Fans: } \% \times (\text{transmission} + \text{product} + \text{infiltration} + \text{internal}); \text{ Defrost: } \text{kW} \cdot \text{cycles} \cdot \text{min}/60 \cdot \text{fraction to room}$$

Item	Inputs	Q [kWh/day]
Fans	8 % of base load	94.1
Defrost	60 kW × 3 × 30 min × 30 %	27.0

Result: **121.11 kWh/day**

Design refrigeration capacity

$$Q_{\text{design}} = \Sigma \text{ loads} \times (1 + \text{safety}) \div \text{run time}$$

Quantity	Value	Unit
Σ loads (subtotal)	1,297.5	kWh/day
Safety / design allowance	10 % = 129.8	kWh/day
Total	1,427.3	kWh/day
Run time	20.0	h/day
Total ÷ run time	71.36	kW
Design capacity	71.36	kW
Evaporator TD / SST	7.0 K / -32.0 °C	

Result: **71.36 kW****Fruit store CS-01****Transmission load**

$$Q = U \cdot A \cdot \Delta T, U = 1 / (1/h_i + L/k + 1/h_o + R_{\text{extra}}), \Delta T = T_{\text{out}} (+ \text{sun effect}) - T_{\text{room}}$$

Surface	A [m²]	U [W/m²K]	T _{out} [°C]	ΔT [K]	Q [kW]	Q [kWh/day]
Wall 1 (length)	270.0	0.214	47.0	46.0	2.653	63.7
Wall 2 (width)	180.0	0.214	47.0	46.0	1.769	42.5
Wall 3 (length)	270.0	0.214	47.0	46.0	2.653	63.7
Wall 4 (width)	180.0	0.214	47.0	46.0	1.769	42.5
Ceiling / roof	600.0	0.214	47.0	46.0	5.896	141.5
Floor	600.0	0.283	12.0	11.0	1.865	44.7

 $h_i = 9.4 \text{ W/m}^2\text{K}$; $h_o = 34 \text{ W/m}^2\text{K}$ outdoors, $9.4 \text{ W/m}^2\text{K}$ to adjacent spaces, none to ground. $T_{\text{room}} = 1.0 \text{ °C}$.

Heat loss to colder surroundings is credited (negative loads reduce the total).

Result: **398.50 kWh/day****Product, packaging and respiration**

$$q = c_{\text{pa}}(T_1 - T_f) + h_{\text{if}} + c_{\text{pb}}(T_f - T_2); c_{\text{pa}} = 3.35 \cdot x_w + 0.84, c_{\text{pb}} = 1.26 \cdot x_w + 0.84, h_{\text{if}} = 334 \cdot x_w \text{ [kJ/kg]} \text{ (unless entered)}; Q = m \cdot q / 3600 \times 24 / t_{\text{pull}} \div \text{CRF};$$

$$Q_{\text{pack}} = m_{\text{pack}} \cdot c_p \cdot \Delta T / 3600 \times 24 / t_{\text{pull}}; Q_{\text{resp}} = (\text{stored}[t] \times w + \text{intake}[t/\text{day}] \times w_{\text{in}}) \times 24 / 1000$$

Product	m [kg/day]	T ₁ →T ₂ [°C]	T _f [°C]	c _{pa} / c _{pb} [kJ/kg·K]	h _{if} [kJ/kg]	q [kJ/kg]	t _{pull} [h]	CRF	Product [kWh/day]	Pack. [kWh/day]	Resp. [kWh/day]
Apples	60,000	22.0 → 1.0	-1.1	3.65 / 1.90	280.6	76.7	24.0	1.00	1,278.9	87.5	

Specific and latent heats from water content (Siebel; latent heat assumes all water freezes) unless entered values are shown as overridden.

Capacity basis “daily”: load concentrated into the pull-down time, expressed per 24 h, then divided by run time with the other loads.

Result: **1,558.40 kWh/day****Door infiltration — Gosney & Olama**

$$q = 0.221 \cdot A \cdot (h_i - h_r) \cdot \rho_r \cdot (1 - \rho_i/\rho_r)^{0.5} \cdot (g \cdot H)^{0.5} \cdot F_m, F_m = [2 / (1 + (\rho_r/\rho_i)^{(1/3)})]^{1.5}; Q = q \cdot t_{\text{open}} \cdot D_f \cdot (1 - E)$$

Door	A [m²]	h _i / h _r [kJ/kg]	ρ _i / ρ _r [kg/m³]	q open [kW]	t _{open} [h/day]	D _f	E	Q [kWh/day]	Moisture [kg/day]
Loading door	7.20	19.7 / 10.2	1.251 / 1.283	16.85	0.611	1.10	0.85	1.7	0.63

 $t_{\text{open}} = (\text{passages} \times \text{open-close time} + 60 \times \text{minutes standing open}) / 3600$.

Air properties at site pressure (ASHRAE psychrometric equations).

Result: **1.70 kWh/day****Internal loads**

$$\text{People: } n \cdot (272 - 6 \cdot t) \text{ W} \cdot \text{h}; \text{Lighting: } \text{W/m}^2 \cdot A_{\text{floor}} \cdot \text{h}; \text{Forklifts: } n \cdot \text{kW} \cdot \text{h}; \text{Other: } \text{kW} \cdot \text{h}$$

Item	Inputs	Q [kWh/day]
People	2 × 266 W × 8 h	4.3
Lighting	8 W/m² × 600 m² × 12 h	57.6
Forklifts	1 × 6 kW × 4 h	24.0
Other equipment	0 kW × 0 h	0.0

Result: **85.86 kWh/day****Evaporator fans and defrost**

$$\text{Fans: } \% \times (\text{transmission} + \text{product} + \text{infiltration} + \text{internal}); \text{Defrost: } \text{kW} \cdot \text{cycles} \cdot \text{min}/60 \cdot \text{fraction to room}$$

Item	Inputs	Q [kWh/day]
Fans	8 % of base load	163.6
Defrost	0 kW × 4 × 30 min × 30 %	0.0

Result: **163.56 kWh/day**

Design refrigeration capacity

$$Q_{\text{design}} = \Sigma \text{ loads} \times (1 + \text{safety}) \div \text{run time}$$

Quantity	Value	Unit
Σ loads (subtotal)	2,208.0	kWh/day
Safety / design allowance	10 % = 220.8	kWh/day
Total	2,428.8	kWh/day
Run time	18.0	h/day
Total ÷ run time	134.93	kW
Design capacity	134.93	kW
Evaporator TD / SST	4.5 K / -3.5 °C	

Result: **134.93 kW****Anteroom / dock AR-01****Transmission load**

$$Q = U \cdot A \cdot \Delta T, U = 1 / (1/h_i + L/k + 1/h_o + R_{\text{extra}}), \Delta T = T_{\text{out}} (+ \text{sun effect}) - T_{\text{room}}$$

Surface	A [m²]	U [W/m²K]	T _{out} [°C]	ΔT [K]	Q [kW]	Q [kWh/day]
Wall 1 (length)	280.0	0.265	47.0	42.0	3.118	74.8
Wall 2 (width)	70.0	0.265	47.0	42.0	0.779	18.7
Wall 3 (length)	280.0	0.265	47.0	42.0	3.118	74.8
Wall 4 (width)	70.0	0.265	47.0	42.0	0.779	18.7
Ceiling / roof	400.0	0.265	47.0	42.0	4.454	106.9
Floor	400.0	0.283	12.0	7.0	0.791	19.0

h_i = 9.4 W/m²K; h_o = 34 W/m²K outdoors, 9.4 W/m²K to adjacent spaces, none to ground.T_{room} = 5.0 °C.

Heat loss to colder surroundings is credited (negative loads reduce the total).

Result: **312.93 kWh/day****Product, packaging and respiration**

$$q = c_{\text{pa}}(T_1 - T_f) + h_{\text{if}} + c_{\text{pb}}(T_f - T_2); c_{\text{pa}} = 3.35 \cdot x_w + 0.84, c_{\text{pb}} = 1.26 \cdot x_w + 0.84, h_{\text{if}} = 334 \cdot x_w \text{ [kJ/kg]} \text{ (unless entered); } Q = m \cdot q / 3600 \times 24 / t_{\text{pull}} \div \text{CRF};$$

$$Q_{\text{pack}} = m_{\text{pack}} \cdot c_p \cdot \Delta T / 3600 \times 24 / t_{\text{pull}}; Q_{\text{resp}} = (\text{stored}[t] \times w + \text{intake}[t/\text{day}] \times w_{\text{in}}) \times 24 / 1000$$

Product	m [kg/day]	T ₁ →T ₂ [°C]	T _f [°C]	c _{pa} / c _{pb} [kJ/kg·K]	h _{if} [kJ/kg]	q [kJ/kg]	t _{pull} [h]	CRF	Product [kWh/day]	Pack. [kWh/day]	Resp. [kWh/day]
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Specific and latent heats from water content (Siebel; latent heat assumes all water freezes) unless entered values are shown as overridden.

Capacity basis “daily”: load concentrated into the pull-down time, expressed per 24 h, then divided by run time with the other loads.

Result: **0.00 kWh/day****Door infiltration — Gosney & Olama**

$$q = 0.221 \cdot A \cdot (h_i - h_r) \cdot \rho_r \cdot (1 - \rho_i/\rho_r)^{0.5} \cdot (g \cdot H)^{0.5} \cdot F_m, F_m = [2 / (1 + (\rho_r/\rho_i)^{1/3})]^{1.5}; Q = q \cdot t_{\text{open}} \cdot D_f \cdot (1 - E)$$

Door	A [m²]	h _i / h _r [kJ/kg]	ρ _i / ρ _r [kg/m³]	q open [kW]	t _{open} [h/day]	D _f	E	Q [kWh/day]	Moisture [kg/day]
Dock levellers ×6 (sealed)	8.40	75.5 / 15.2	1.094 / 1.264	271.70	2.000	0.80	0.90	43.5	17.77

t_{open} = (passages × open-close time + 60 × minutes standing open) / 3600.

Air properties at site pressure (ASHRAE psychrometric equations).

Result: **43.47 kWh/day****Internal loads**

$$\text{People: } n \cdot (272 - 6 \cdot t) \text{ W} \cdot \text{h}; \text{Lighting: } \text{W/m}^2 \cdot A_{\text{floor}} \cdot \text{h}; \text{Forklifts: } n \cdot \text{kW} \cdot \text{h}; \text{Other: } \text{kW} \cdot \text{h}$$

Item	Inputs	Q [kWh/day]
People	8 × 242 W × 16 h	31.0
Lighting	8 W/m² × 400 m² × 12 h	38.4
Forklifts	3 × 6 kW × 12 h	216.0
Other equipment	0 kW × 0 h	0.0

Result: **285.38 kWh/day****Evaporator fans and defrost**

$$\text{Fans: } \% \times (\text{transmission} + \text{product} + \text{infiltration} + \text{internal}); \text{Defrost: } \text{kW} \cdot \text{cycles} \cdot \text{min}/60 \cdot \text{fraction to room}$$

Item	Inputs	Q [kWh/day]
Fans	8 % of base load	51.3
Defrost	0 kW × 4 × 30 min × 30 %	0.0

Result: **51.34 kWh/day**

Design refrigeration capacity

$$Q_{\text{design}} = \Sigma \text{ loads} \times (1 + \text{safety}) \div \text{run time}$$

Quantity	Value	Unit
Σ loads (subtotal)	693.1	kWh/day
Safety / design allowance	10 % = 69.3	kWh/day
Total	762.4	kWh/day
Run time	18.0	h/day
Total $\div$ run time	42.36	kW
Design capacity	42.36	kW
Evaporator TD / SST	8.0 K / -3.0 °C	

Result: **42.36 kW**

6. Load summary

Load component	Frozen store FS-01	Fruit store CS-01	Anteroom / dock AR-01	Total [kWh/day]	Share
Transmission	690	399	313	1,401	33.4 %
Product	197	1,279	0	1,476	35.2 %
Packaging	7	88	0	95	2.3 %
Respiration	0	192	0	192	4.6 %
Infiltration	12	2	43	57	1.4 %
Ventilation air	0	0	0	0	0.0 %
People	7	4	31	42	1.0 %
Lighting	96	58	38	192	4.6 %
Motors / forklifts	168	24	216	408	9.7 %
Equipment	0	0	0	0	0.0 %
Fans	94	164	51	309	7.4 %
Defrost	27	0	0	27	0.6 %
Calculated load (Σ)	1,298	2,208	693	4,199	100 %

7. Design allowances

Room	Σ loads [kWh/day]	Safety [%]	Allowance [kWh/day]	Total [kWh/day]	Run time [h/day]	Product basis	Pull-down adj. [kW]
Frozen store FS-01	1,298	10	130	1,427	20	daily	-
Fruit store CS-01	2,208	10	221	2,429	18	daily	-
Anteroom / dock AR-01	693	10	69	762	18	daily	-

8. Final refrigeration load

Room	T [°C]	SST [°C]	Design capacity
Frozen store FS-01	-25.0	-32	71.4 kW
Fruit store CS-01	1.0	-3	134.9 kW
Anteroom / dock AR-01	5.0	-3	42.4 kW

Suction level	Rooms	Capacity
-3 °C	Fruit store CS-01, Anteroom / dock AR-01	177.3 kW
-32 °C	Frozen store FS-01	71.4 kW
-42 °C	Blast freezer BF-01	134.1 kW

Tunnel / blast freezer	Air [°C]	Freezing time [h]	SST [°C]	Design capacity
Blast freezer BF-01	-35.0	11.5	-42	134.1 kW

Total design refrigeration load (rooms + tunnels): **382.8 kW**

9. Tunnel / blast freezers

Blast freezer BF-01 — batch

Input	Value
Product	Poultry, fresh (tabulated library)
Shape / D	Slab / carton (D = thickness) / 0.100 m
h _{air} / R _{pack}	20.0 W/m ² K / 0.010 m ² K/W
Air / inlet / centre / final	-35.0 / 5.0 / -18.0 / -18.0 °C
Batch / loading time	12,000 kg / 1.00 h
Freezing-time basis	pham
Loss & safety	10 % (Q ÷ (1 - x))

Freezing time

Plank: $t = \rho_f \cdot L / (T_f - T_m) \cdot (P \cdot D / h + R \cdot D^2 / k_f)$. Pham: $t = d / (E \cdot h) \cdot (\Delta H_1 / \Delta T_1 + \Delta H_2 / \Delta T_2) \cdot (1 + Bi/2)$; $T_{fm} = 1.8 + 0.263 \cdot T_c + 0.105 \cdot T_m$; $h = 1 / (1/h_{air} + R_{pack})$

Quantity	Value	Unit
Shape (P, R, E)	Slab / carton (D = thickness) (0.500, 0.1250, 1)	
D / d	0.100 / 0.050	m
$h_{air} \rightarrow h_{eff}$	20.0 $\rightarrow$ 16.67	W/m²K
k_f, ρ_u, ρ_f	1.40, 1,050, 1,000	W/mK, kg/m³
Biot number $h \cdot d / k_f$	0.595	-
$T_i / T_f / T_c / T_m$	5.0 / -1.0 / -18.0 / -35.0	°C
T_{fm} (mean freezing temperature)	-6.61	°C
$\Delta H_1 / \Delta H_2$	40.82 / 267.50	MJ/m³
$\Delta T_1 / \Delta T_2$	34.20 / 28.39	K
Plank freezing time	7.86	h
Pham freezing time	11.48	h
Design basis	Pham	

Plank ignores pre-cooling above and sub-cooling below the freezing point, so it underestimates the time; Pham includes both.

Result: **11.48 h**

Product load (continuous-flow method)

$\dot{m} = \text{batch} / t_{\text{freeze (batch)}}$ or throughput (continuous); $Q = \dot{m} \cdot [c_1(t_1 - t_f) + h_{if} + c_2(t_f - t_2)] / 3600 \times \text{load factor}$

Quantity	Value	Unit
Product	Poultry, fresh (library)	
$c_1 / c_2 / h_{if}$	3.349 / 1.800 / 247.0	kJ/kg·K, kJ/kg
$t_1 \rightarrow t_2$ (t_f)	5.0 $\rightarrow$ -18.0 (-1.0)	°C
Heat removed q	297.7	kJ/kg
Product flow $\dot{m}$	1,045.3	kg/h
Above freezing / latent / below freezing	5.83 / 71.72 / 8.89	kW
Load distribution factor	1.00	-

Result: **86.44 kW**

Other tunnel loads

Packaging $\dot{m} \cdot c_p \cdot \Delta T$; trays/trolleys per batch; transmission $U \cdot A \cdot \Delta T$; infiltration Gosney–Olama $\times$ open time $\times D_f 0.8 \times (1 - E)$; fans, lights, other, defrost as entered

Item	Value	Unit
Packaging	0.45	kW
Trays / trolleys / racks	0.00	kW
Transmission	1.58	kW
Door infiltration	1.72	kW
Fans	30.00	kW
Lighting	0.50	kW
Other equipment	0.00	kW
Defrost heat	0.00	kW
Walls + ceiling area / U	216.0 m² / 0.107	W/m²K
Floor area / U	72.0 m² / 0.192	W/m²K
Loading door	q open 370.0 kW, 0.067 h/cycle, E 0.00	

Result: **34.25 kW**

Tunnel design refrigeration capacity

$Q_{\text{design}} = \Sigma \text{loads} \div (1 - x)$ (workbook loss & safety method)

Quantity	Value	Unit
Σ loads	120.69	kW
Loss & safety	10 % $\rightarrow$ 13.41	kW
Design capacity	134.10	kW
Air temperature / TD / SST	-35.0 / 7.0 / -42.0	°C, K, °C
Air volume for ΔT_{air}	108,026 (ΔT 3.0 K)	m³/h
Specific energy	462 kJ/kg (110 kcal/kg)	
Throughput	23,078 kg/day	

Result: **134.10 kW**

10. Machinery room ventilation**Engine room ER-01 — ANSI/IIAR 2-2008 Addendum A (2010-)**

Common standard: Ammonia machinery room — IIAR 2 v2; overridden fields: Detector setpoint (ppm)

Mode	Basis	m ³ /h	L/s	cfm	ACH
Normal	20 ACH	63,000	17,500	37,080	20.0
Continuous	—	0	0	0	0.0
Emergency	30 ACH (one air change every 2 min)	94,500	26,250	55,621	30.0

Room heat: 34.4 kW (motors 33.3, envelope 1.1); volume 3,150 m³.

- Ambient design above 37.2 °C (99 °F): IIAR 2 permits the emergency ventilation system to supplement normal ventilation during extreme conditions.
- Δ Supply air is at or above the 40 °C room limit, so temperature control by ventilation alone is not possible: provide room cooling or electrical equipment rated above 40 °C (IIAR 2 exception).
- At least two ammonia detectors: one at $\leq$ TLV-TWA (25 ppm) to start normal ventilation at full design capacity and alarm; one at $\leq$ 1000 ppm to start emergency ventilation (IIAR 2).
- Alarm to a monitored location; visual and audible alarms inside the machinery room and outside each entrance.
- Failure of any single fan must not reduce total ventilation below 20 ACH.
- Exhaust fans discharge vertically upward at $\geq$ 12.7 m/s (2500 fpm) with non-sparking blades.
- Make-up air keeps the room negative, not more than 62 Pa (0.25 in. w.c.).
- Ventilation operable manually as well as by detectors and thermostat; emergency ventilation switch outside the principal entrance.

11. Engineering notes and validation

Level	Location	Message
INFO	Frozen store FS-01	Room below 0 °C: provide under-floor heating/ventilation against frost heave and set the ground temperature to the heater set-point.

12. Standards and references

Reference	Edition	Clause	Classification	Application
ANSI/ASHRAE 15 — Safety Standard for Refrigeration Systems	To be verified by user	—	Required / code-based	Halocarbon / non-ammonia machinery rooms
ANSI/IIAR 2	2008 Addendum A (2010)	13.2.1.1-13.2.1.2	Required / code-based	Alarm to monitored location; visual + audible alarms inside the room and outside each entrance
ANSI/IIAR 2	2008 Addendum A (2010)	13.2.3.1-13.2.3.2	Required / code-based	Detector 1 ≤ TLV-TWA starts normal ventilation at full capacity + alarm; detector 2 ≤ 1000 ppm starts emergency ventilation + alarm
ANSI/IIAR 2	2008 Addendum A (2010)	13.3.1	Required / code-based	Mechanical ventilation to outdoors, actuated by detectors, temperature and manually; powered independently of machinery; not subject to emergency shutdown
ANSI/IIAR 2	2008 Addendum A (2010)	13.3.2	Required / code-based	Failure of any single fan shall not reduce total ventilation below 20 ACH
ANSI/IIAR 2	2008 Addendum A (2010)	13.3.3	Required / code-based	Make-up air; room negative, max 0.25 in. w.c. (≈ 62 Pa)
ANSI/IIAR 2	2008 Addendum A (2010)	13.3.7.1-13.3.7.2	Required / code-based	Exhaust discharges vertically ≥ 2500 fpm (12.7 m/s); non-sparking fan blades
ANSI/IIAR 2	2008 Addendum A (2010)	13.3.8.1	Required / code-based	Normal ventilation = greater of 20 ACH or flow limiting room to 40 °C (104 °F) with 1 % ASHRAE design inlet air; emergency fans may supplement where ambient can exceed 37.2 °C (99 °F)
ANSI/IIAR 2 — Safe Design of Closed-Circuit Ammonia Refrigeration Systems	2008 Addendum A (2010)	13.3.9.1	Required / code-based	Emergency ventilation ≥ 30 ACH on gross room volume
ANSI/IIAR 2 (current edition)	To be verified by user	—	Required / code-based	Design, equipment, machinery rooms for closed-circuit ammonia systems
ANSI/IIAR 9 — Minimum Safety Criteria for a Safe Ammonia Refrigeration System	To be verified by user	—	Required / code-based	Safety criteria for existing systems
ASHRAE Handbook—Fundamentals, chapter "Climatic Design Information"	To be verified by user	—	Recommended practice	Outdoor design dry-bulb (0.4 %/1 %) and mean coincident wet-bulb
ASHRAE Handbook—Fundamentals, chapter "Psychrometrics"	To be verified by user	—	Recommended practice	Saturation pressure (Hyland-Wexler), humidity ratio, enthalpy, specific volume
ASHRAE Handbook—Refrigeration, chapter "Cooling and Freezing Times of Foods"	To be verified by user	—	Recommended practice	Plank equation and shape factors; Pham (1986) method with mean freezing temperature $T_{fm} = 1.8 + 0.263 \cdot T_c + 0.105 \cdot T_m$; surface heat-transfer coefficients
ASHRAE Handbook—Refrigeration, chapter "Refrigerated-Facility Loads"	To be verified by user	—	Recommended practice	Transmission, sun effect, product, infiltration (Gosney-Olama), internal and equipment loads, safety factor, run time
ASHRAE Handbook—Refrigeration, chapter "Thermal Properties of Foods"	To be verified by user	—	Recommended practice	Water content, freezing point, respiration heat; Siebel specific-heat equations
Empirical air-change formulas (user-supplied heat-load workbook)	—	—	Recommended practice	Storage rooms $n = 70/\sqrt{V} \times f$ per day; docks/manipulation rooms $n = 35/\sqrt{V} \times f_n$ per hour; manual minimum 2/day
EN 378 — Refrigerating systems and heat pumps: safety and environmental requirements	To be verified by user	—	Required / code-based	European projects
Freezing of goods — continuous-flow method (user-supplied heat-load workbook)	—	—	Recommended practice	$\dot{m} = \text{batch} / \text{freezing time}$; $Q = \dot{m} \cdot [C_1(t_1 - t_o) + h + C_2(t_o - t_2)]$; loss & safety $Q \div (1 - x)$; tabulated product properties (kcal → kJ, 4.186)
Gosney & Olama door-infiltration equation	As presented in ASHRAE Handbook—Refrigeration	—	Recommended practice	Refrigeration load through an open door with D_t , D_f and protection effectiveness E
IIAR Machinery Room Ventilation Analysis Tool	v2.29	—	Recommended practice	Rate formulas for IIAR 2 (1992/1999/2008A), IMC, ASHRAE 15, UMC/CMC used by the ventilation module
ISO 5149 — Refrigerating systems and heat pumps: safety and environmental requirements	To be verified by user	—	Required / code-based	International projects
Plank freezing-time equation	As presented in ASHRAE Handbook—Refrigeration, Stoecker and Dossat	—	Recommended practice	$t = \rho \cdot L / \Delta T \cdot (P \cdot D / h + R \cdot D^2 / k)$; slab $P \frac{1}{2} R \frac{1}{3}$, cylinder $\frac{1}{4} R \frac{1}{16}$, sphere $\frac{1}{6} R \frac{1}{24}$
R. J. Dossat, Principles of Refrigeration	To be verified by user	—	Recommended practice	Air changes per 24 h vs room volume and usage factors; product, occupancy and miscellaneous loads; chilling rate factor
W. F. Stoecker, Industrial Refrigeration Handbook	McGraw-Hill (edition to be verified by user)	—	Recommended practice	Refrigeration load components, door infiltration, industrial practice

Clause numbers are listed only where verified from a supplied document. An internal approval status does not constitute regulatory approval.

13. Revision history

Rev	Date	By	Description	Status	Total load
00	2026-09-25	QA Engineer	Migrated from v1 localStorage (original revision "A")	Draft	235.8 kW
01	2026-09-25	QA Engineer	QA revision	Issued	382.8 kW

14. Prepared / Checked / Approved

	Name	Date	Signature
Prepared	QA Engineer	2026-09-25	
Checked			
Approved			

Review status reflects internal engineering review only and does not imply approval by any authority having jurisdiction.